ENGINEERS WALLAH

CHEMISTRY

Lecture II

Matter and Its Classification - II

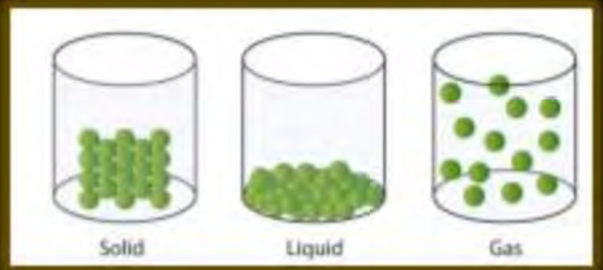


By: Sunil Bhaiya

TOPICS *to be covered*

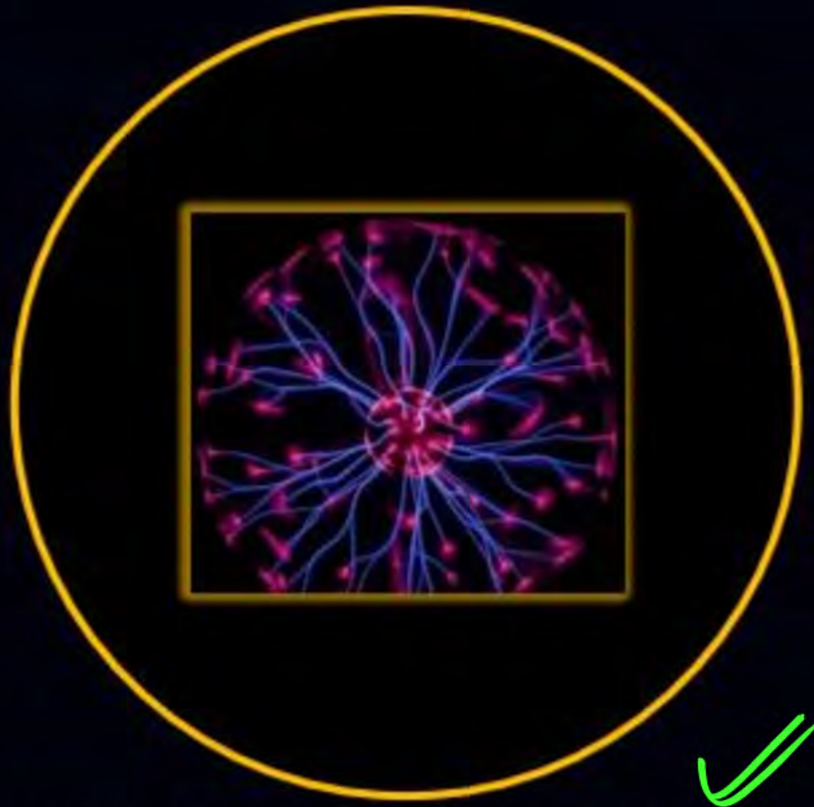
1. Atom and Basic Terminologies
2. Introduction to Plasma and BEC
3. Bohr's-Bury Rule/Scheme

Knowledge Ride On



Atom and Basic Terminologies

Knowledge Ride On



Introduction to Plasma and BEC

Knowledge Ride On



Bohr's-Bury Rule/Scheme

Knowledge Ride On



Efficiency Hacks



Glossary

Atom (परमाणु)
Indivisible (अविभाज्य)
Nucleus(नाभिक)
Element (तत्व)
Valence shell (संयोजकता कोश)
Penultimate shell (उपांतिम कोश)
Kinetic energy (गतिज ऊर्जा)
Heat energy (ऊष्मीय ऊर्जा)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

The state in which inter-particle force of attraction is very strong is:

RRB JE 2014

Order of interparticle f.o.a.

Solids > Liquids > Gases

A Solid

B Liquid

C Gas

D Vapour

form of gas

Question

Which among the following set of conditions favours liquefaction of gases?

RRB JE 2015

A Low pressure and low temperature

B High pressure and high temperature

C High pressure and low temperature

D Low pressure and high temperature

Question

Connversion of a solid directly to gaseous state is known as _____.

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A Sublimation

B Evaporation

C Diffusion

D Liquefaction

Concept Bites – Atom and Basic Terminologies

Yeh Atom Atom Kya Hai??



A- TOM → Nahi

Basic Introduction to Atom

Dalton

C-I

Derived from Greek word 'atomos' which means uncuttable/indivisible.

C-II

Smallest particle of any substance that may or may not be capable of free existence.



Democritus (Around 400 BC)

Father of Early Atomic Theory

Maharishi Kanad (Around 600 BC)

Father of Indian Atomic Theory

Basic Introduction to Atom

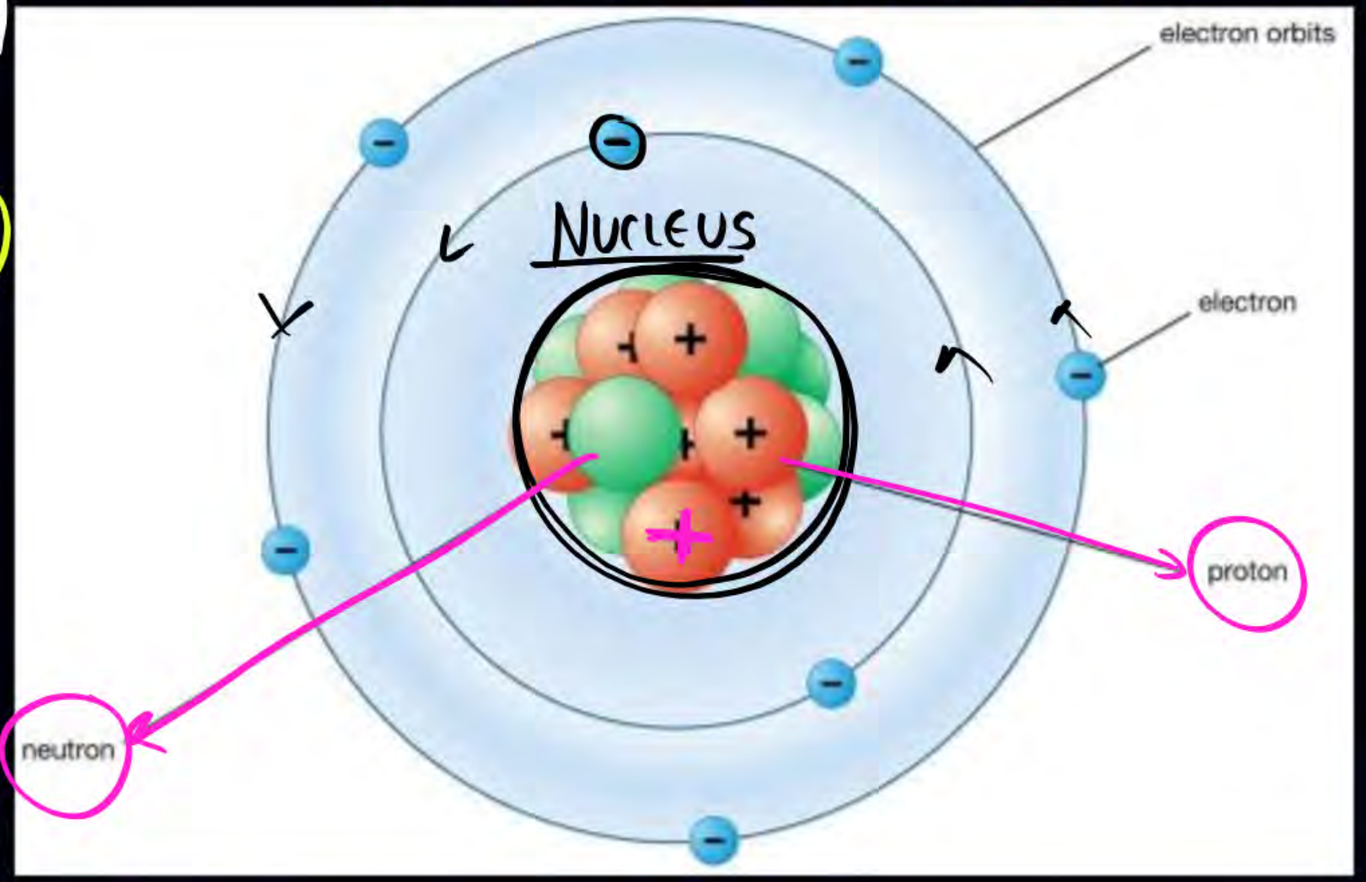
Atom is divisible → made from sub-atomic particles.

* Nucleus → discovered by Rutherford

(+) Nucleus has:
- (a) +vely charged particle (PROTON)
- (b) Neutral particle (NEUTRON)

(*) Extranuclear part:
- (a) Electrons (negatively charged particle)

Bohr's Atomic Model



Characteristics of Sub-atomic Particles

**

Sub-atomic Particle	Mass	Charge	Discovered By
Electron	9.1×10^{-31} kg	Negatively charged	J.J. Thomson
Proton	1.67×10^{-27} kg	Positively charged	E. Rutherford
Neutron	1.67×10^{-27} kg	Neutral	J. Chadwick

E. Goldstein

Atomic Number (Z)

C-I

It is the number of protons present in the nucleus of an atom.

C-II

Atom is electrically neutral (→ overall no charge) because:

no. of electrons = no. of protons



Characteristics of Atomic Number

no fractional values

1 ✓ 1.5 ✗
2 ✓ 2.6 ✗
3 ✓ 3.8 ✗

- It is always a whole number quantity.
- No two elements can have the same atomic number.
- Atomic number of an element always remain the same.
- All atoms of the same element have the same atomic number.

Trick to Remember First 20 Elements













Oxygen (O) 8

Fluorine (F) 9

Neon (Ne) 10

Hey Henry Listen B B.C. News On Friday Night

Hydrogen (H) 1

Helium (He) 2

Lithium (Li) 3

Beryllium (Be) 4

Boron (B) 5

Carbon (C) 6

Nitrogen (N) 7

A.No. (Z)

Trick to Remember First 20 Elements



Nariyal
↓
Sodium (Na)
(11)



Mango
↓
(12) Magnesium (Mg)



Aloo
(13) Aluminium (Al)

(14) Silicon (Si)



Salad

(15) Phosphorus (P)



Pepsi

Soda
↓
(16) Sulphur (S)



Coca-Cola
↓
(17) Chlorine (Cl)



Aur → Argon (Ar) (18)



Potassium (K) (19)

Kaju - Katli → Calcium (Ca) (20)



Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following element has 12 as its atomic number?

SSC JE 2020

A Potassium (19)

B Magnesium (12)

C Aluminium (13)

D Sodium (11)

Question

Which of the following element has 20 as its atomic number?

A Potassium (19)

B Aluminium (13)

C Calcium (20)

D Sodium (11)

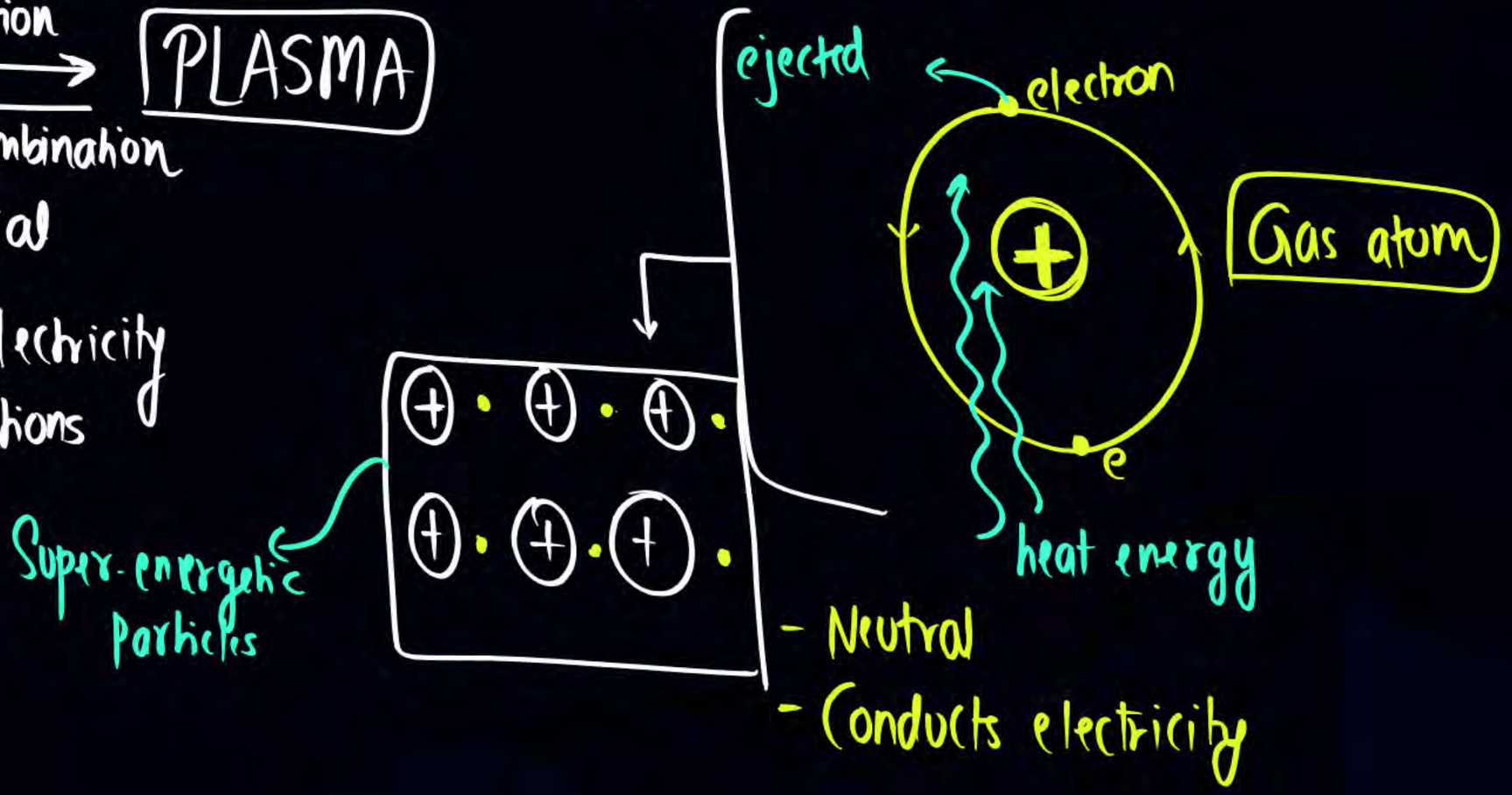
Concept Bites – Introduction to Plasma and BEC

Introduction to Plasma

- 4th state of matter
- On Earth → [- Northern Lights (Aurora Boriolis) - Lightning]
- Sun, Stars → 'PLASMA'

GAS ⇄ (Ionisation / Recombination) PLASMA

- Electrically neutral
- Don't conduct electricity in normal conditions



Introduction to BEC

→ Bose-Einstein Condensate

* fifth state of matter

1920 - Satyendra Bose & Albert Einstein → first prediction

1995 - Proved that it exists

* Low density gas
↓
Extremely cool
↓
motion of atoms of gases (CEASE)

→ highly unexcited particles → B.E.C.

Order of Kinetic Energy (गतिज ऊर्जा) in States of matter

energy associated with motion of particles

B.E.C < Solids < Liquids < Gas < Plasma

B.E.C → Kinetic energy ≈ 0

Solids: ○○○

Liquids: ○ ○

Gas: ○ ○ ○

Concept Bites – Bohr's-Bury Rule/Scheme

Neil's Bohr
Charles Bury
1921

Bohr's-Bury Rule/Scheme

(i) The maximum number of electrons that can be accommodated in a shell is given by the formula $2n^2$. Moreover, filling of electrons always start from inner shells.

Shell / energy level / index / orbit



innermost orbit → K, $n=1$, $2\times(1)^2 = 2$ electrons

L, $n=2$, $2\times(2)^2 = 8$ electrons

M, $n=3$, $2\times(3)^2 = 18$ electrons

N, $n=4$, $2\times(4)^2 = 32$ electrons

⋮

Electronic Configuration of Lithium

Element	Atomic or proton number	Number of electrons
Lithium	3	3

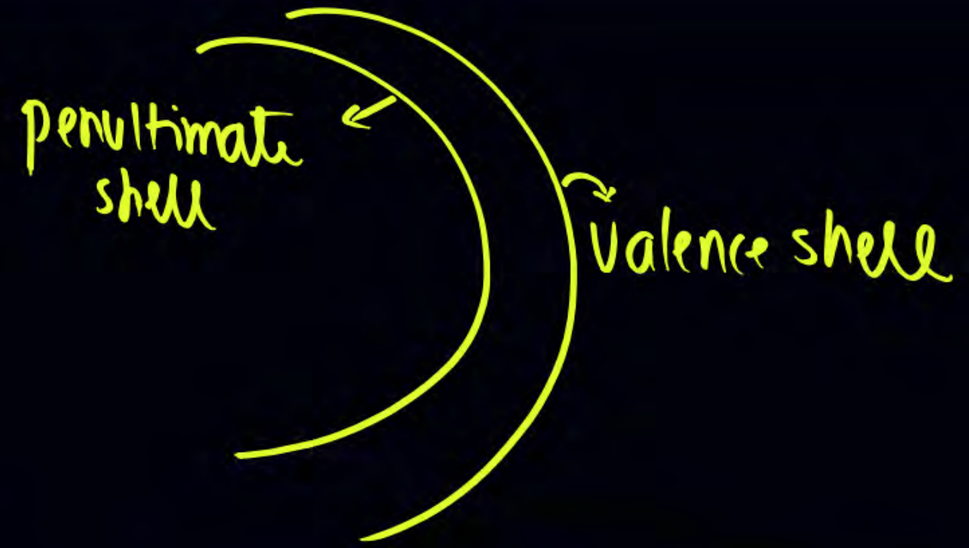
→ K L
2 1

	A.No.(Z)	no. of electrons
Hydrogen (H)	1	1
Helium (He)	2	2

K	L
1	
2	

Bohr's-Bury Rule/Scheme

(ii) The outermost shell/valence shell (संयोजकता कोश) cannot have more than 8 electrons and the next inner shell to it, i.e. penultimate shell (उपांतिम कोश) cannot have more than 18 electrons.



Electronic Configuration of Lithium

Element	Atomic or proton number	Number of electrons
Krypton (Kr)	36	36



	A.No.(Z)	no. of electrons	K	L	M
→ Carbon (C)	6	6	2	4	
→ Neon (Ne)	10	10	2	8	
→ Sodium (Na)	11	11	2	8	1
→ Chlorine (Cl)	17	17	2	8	7

Bohr's-Bury Rule/Scheme

(iii) Even if the capacity of the penultimate shells exceeds 8, it cannot have more than 8 electrons unless there are 2 electrons in the outermost shell./valence shell

	A.No.(Z)	no. of electrons	K	L	M	N	
Potassium (K)	19	19	2	8	9		(X)
			2	8	8	1	(✓)

Electronic Configuration of Lithium

Element	Atomic or proton number	Number of electrons
Calcium (a)	20	20

K L M N
2 8 8 2

Scandium (Sc) २१ २१ 2 8 9 2

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

What is the electronic distribution in Aluminium atom?

RRB JE 2019

A. 8, 2, 3

B. 2, 8, 3

C. 2, 8, 2

D. 2, 3, 8

A. No. (Z) = 13

↓

no. of electrons = 13

K	L	M
2	8	3

Question

An atom has electronic configuration 2, 8, 7. What is the atomic number of this element?

RRB JE 2019

Total electrons = 2 + 8 + 7 = 17

Electrically neutral

No. of protons = 17

A. No. (Z) = 17

A 18

B 16

C 17

D 20

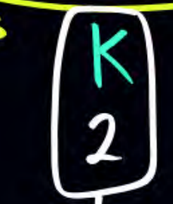
Question

Helium atom has _____ electrons in its outermost shell.

RRB JE 2019

	A. No. (Z)	no. of electrons
Helium (He)	2	2

valence shell/



outermost shell/
valence shell

A. Four

B. Two ✓

C. One

D. Three

Question

Which element has the least number of valence electrons?

→ सबसे कम

RRB JE 2015

electrons in valence shell

		A.No. (Z)	No. of electrons
A	Chlorine (Cl)	17	17
B	Sodium (Na)	11	11
C	Fluorine (F)	9	9
D	Oxygen (O)	8	8

K	L	M	N
2	8	7	
2	8	1	
2	7		
2	6		

Question

Which of the following set of elements has same number of valence electrons?

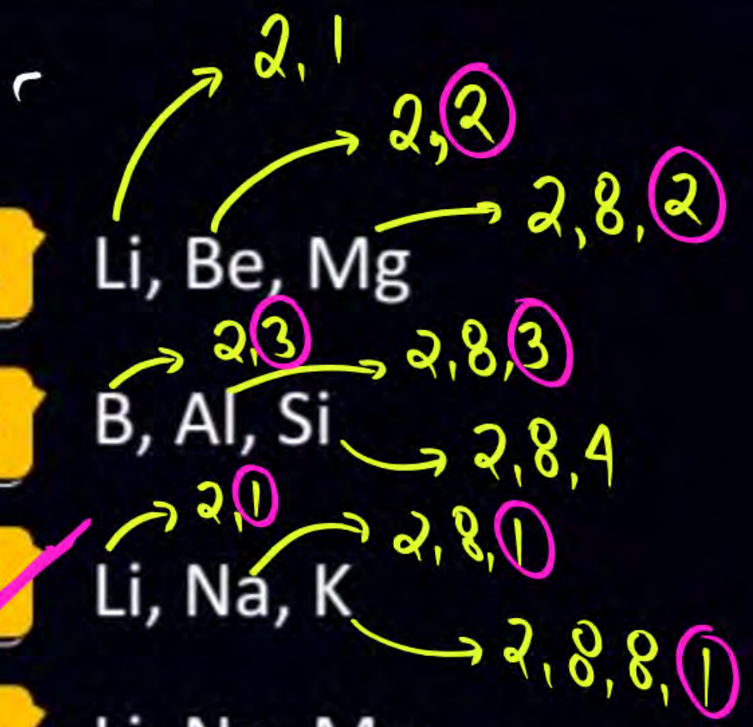
RRB JE 2015

A Li, Be, Mg

B B, Al, Si

C Li, Na, K

D Li, Na, Mg



Efficiency Hacks!

Tips to Avoid Burn Out!

(i) Ideally after every 45 minutes of study take 5-10 minutes break.

(ii) During small breaks; listen to mind calming music and have your supper. Do meditation if possible.




#sbsathhai
#pwsathhai
Thank
You !